

Systematic Trading Strategies with Machine Learning Algorithms

Supervised Learning Algorithms



May 5, 2026

Introducing Ensemble Models

- Decision Trees for Classification and Regression

- Bagging - Random Forest

- Boosting: Adaboost

- Gradient Boosting Algorithm

Introducing Neural Networks

- Shallow Neural Networks - Forward Propagation -

- Activation Functions

- Setting the loss function for Classification and Regression

- Learning the parameters using Gradient Descent

- Introducing the Variable Selection Network

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Algorithm Decision Tree Learning Algorithm

Require: Training data $\{(F_i, y_i)\}_{i=1}^n$, stopping criteria

Ensure: Decision tree T

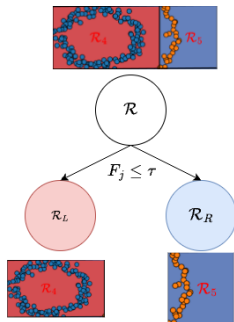
- 1: Initialize tree with single root node containing all data
 - 2: **while** nodes can be split and stopping criteria not met **do**
 - 3: **for** each leaf node with region $\mathcal{R}$ **do**
 - 4: Find (j^*, τ^*) that maximizes:
 - 5:
$$IG(j, \tau) = I(\mathcal{R}) - \frac{|\mathcal{R}_L|}{|\mathcal{R}|} I(\mathcal{R}_L) - \frac{|\mathcal{R}_R|}{|\mathcal{R}|} I(\mathcal{R}_R)$$
 - 6: Where $\mathcal{R}_L = \{F \in \mathcal{R} : F_j \leq \tau\}$ and $\mathcal{R}_R = \{F \in \mathcal{R} : F_j > \tau\}$
 - 7: Split node using rule $F_{j^*} > \tau^*$
 - 8: **end for**
 - 9: **end while**
 - 10: Assign prediction to each leaf node (majority class)
 - 11: **return** T
-

- **Information Gain** for feature j and threshold τ :

$$IG(j, \tau) = I(\mathcal{R}) - \frac{|\mathcal{R}_L|}{|\mathcal{R}|} I(\mathcal{R}_L) - \frac{|\mathcal{R}_R|}{|\mathcal{R}|} I(\mathcal{R}_R)$$

Where:

- $\mathcal{R}_L = \{F \in \mathcal{R} : F_j \leq \tau\}$
 - $\mathcal{R}_R = \{F \in \mathcal{R} : F_j > \tau\}$
-
- The DT algorithm:
 1. For each feature j and possible threshold τ , compute $IG(j, \tau)$
 2. Select feature j^* and threshold τ^* that maximize IG
 3. Split node and create child regions $\mathcal{R}_L$ and $\mathcal{R}_R$
 4. Recursively apply to each child node until stopping criteria met



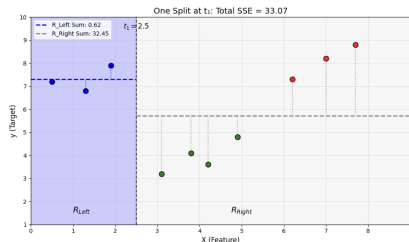
- ▶ For a split at value t_1 on feature X :

- ▶ Left region:

$$R_{Left} = \{x_i | x_i \leq t_1\}$$

- ▶ Right region:

$$R_{Right} = \{x_i | x_i > t_1\}$$



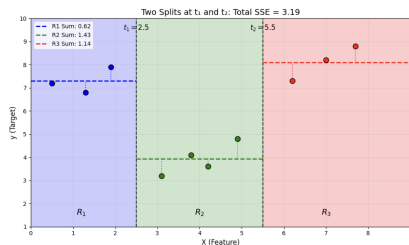
- ▶ For each region, we compute:
 - ▶ Prediction value: average of y_i in the region
 - ▶ SSE: sum of squared errors in the region
- ▶ **Optimization objective:** Choose feature j and threshold t that minimizes:

$$SSE_{total} = SSE_{Left} + SSE_{Right}$$

- ▶ The prediction function for M regions $(R_m)_{1 \leq m \leq M}$ is:

$$f(x) = \sum_{m=1}^M c_m \mathbb{1}\{x \in R_m\}$$

- ▶ Where c_m is the average of all y_i for which $x_i \in R_m$



- ▶ After recursive splitting, we end up with multiple regions (leaves).
- ▶ Increasing the number of regions leads to lower training error
- ▶ To avoid overfitting, we need **stopping criteria**: Maximum depth, minimum samples per leaf, minimum error improvement.

Algorithm Regression Tree Learning Algorithm

Require: Training data $\{(F_i, y_i)\}_{i=1}^n$, stopping criteria

Ensure: Regression tree T

- 1: Initialize tree with single root node containing all data
- 2: **while** nodes can be split and stopping criteria not met **do**
- 3: **for** each leaf node with region $\mathcal{R}$ **do**
- 4: Find (j^*, τ^*) that minimizes:
- 5:
$$SSE(j, \tau) = \sum_{i: F_i \in \mathcal{R}_L} (y_i - \bar{y}_{\mathcal{R}_L})^2 + \sum_{i: F_i \in \mathcal{R}_R} (y_i - \bar{y}_{\mathcal{R}_R})^2$$
- 6: Where $\mathcal{R}_L = \{F \in \mathcal{R} : F_j \leq \tau\}$ and $\mathcal{R}_R = \{F \in \mathcal{R} : F_j > \tau\}$
- 7: Split node using rule $F_{j^*} > \tau^*$
- 8: **end for**
- 9: **end while**
- 10: Assign prediction $\bar{y}_{\mathcal{R}_m}$ to each leaf node (average of y_i in the region)
- 11: **return** T

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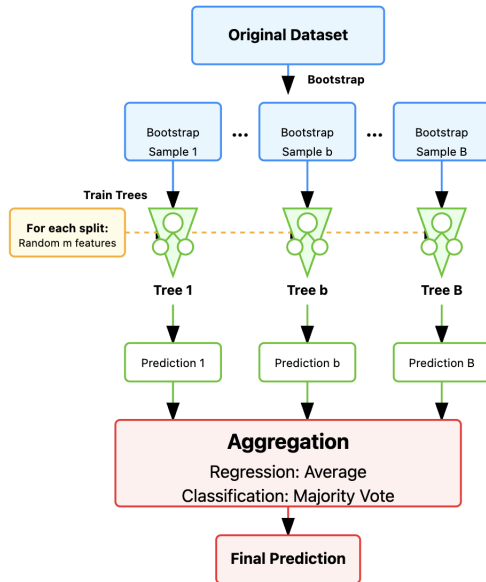
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Introducing the Variable Selection Network

- ▶ **Motivation of Ensemble Models:** Aggregate weak learners to build a strong learner
- ▶ **Bagging Methodology** (Bagging: Bootstrap Aggregation):
 1. **Generate bootstrap samples** $\mathcal{B}_1, \dots, \mathcal{B}_B$:
 - ▶ Create $\mathcal{B}_b$ by picking points from $\{x_1, \dots, x_n\}$ randomly n times
 - ▶ A particular x_i can appear in $\mathcal{B}_b$ many times.
 2. **Train a model per bootstrap:**
 - ▶ Each bootstrap sample trains an independent model
 3. **Aggregate the predictions:**
 - ▶ **Classification:** Majority vote across all models
 - ▶ **Regression:** Average of individual model predictions

- ▶ **Random Forest** is an ensemble method that improves upon bagging decision trees by introducing additional randomness at each split for each decision tree:
 - ▶ During training, at each node, only consider a subset of features
 - ▶ Result: Less correlated trees



Algorithm Random Forest Algorithm

Require: Training data $\{(F_i, y_i)\}_{i=1}^n$, number of trees B , features per split $m < p$

Ensure: Random Forest model RF

```
1: for  $b = 1$  to  $B$  do
2:   Draw a bootstrap sample  $\mathcal{B}_b$  of size  $n$  from the training data
3:   Initialize tree  $T_b$  with root node containing data from  $\mathcal{B}_b$ 
4:   while nodes in  $T_b$  can be split do
5:     for each leaf node with region  $\mathcal{R}$  do
6:       Randomly select  $m$  features from the available  $p$ 
7:       Find best split  $(j^*, \tau^*)$  among these  $m$  features
8:       Split node using rule  $F_{j^*} > \tau^*$ 
9:     end for
10:  end while
11: end for
12: return  $RF = \{T_1, T_2, \dots, T_B\}$ 
```

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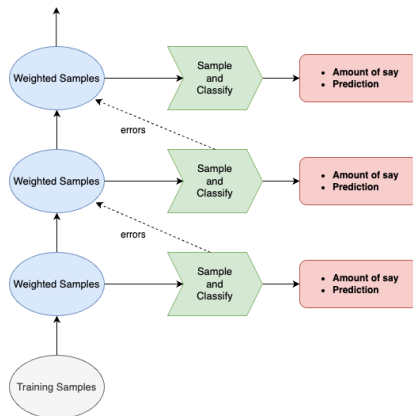
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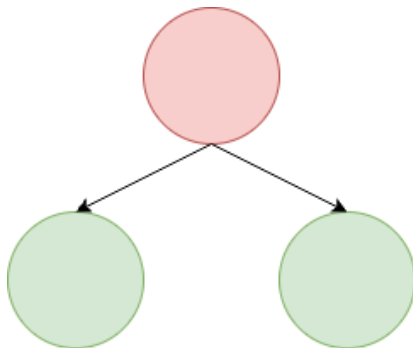
Adaptive Sample Weighting

- ▶ Unlike bagging, AdaBoost [2] does not use uniform sampling
- ▶ Samples are weighted based on classification difficulty
- ▶ Misclassified examples receive higher weights in subsequent iterations
- ▶ The algorithm creates models sequentially, each one focusing on correcting previous errors
- ▶ This adaptive weighting is the core mechanism behind boosting's effectiveness



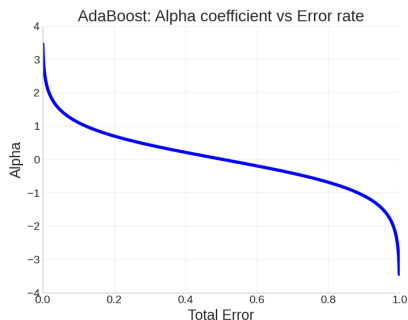
Leveraging Weak Learners (Decision Stumps)

- ▶ AdaBoost builds its power from simple decision trees (typically depth=1) called **stumps**
- ▶ Individually, each model performs only slightly better than random guessing
- ▶ However, they are computationally efficient and resistant to overfitting
- ▶ The algorithm's strength comes from combining many weak models into a strong ensemble



Performance-Based Model Weighting

- ▶ Unlike random forests where all trees contribute equally, AdaBoost assigns varying importance to each model
- ▶ The algorithm calculates an "amount of say" (α_t) for each classifier based on its accuracy
- ▶ Highly accurate classifiers receive strong positive weights
- ▶ Random-level performers (error rate = 0.5) receive zero weight
- ▶ Poor performers can contribute negatively by having their predictions reversed



1. How do we determine each model's contribution?

- ▶ Model influence (α_t) is based on weighted error
$$\epsilon_t = \sum_{i=1}^n w_t(i) \mathbb{1}\{y_i \neq f_t(x_i)\}$$
- ▶ $\alpha_t = \frac{1}{2} \ln\left(\frac{1-\epsilon_t}{\epsilon_t}\right)$ where ϵ_t is the proportion of weighted misclassifications

2. How do we adaptively weight training examples?

- ▶ Weights are updated after each iteration:
$$w_{t+1}(i) \propto w_t(i) \cdot e^{\alpha_t(1-2\mathbb{1}\{y_i=f_t(x_i)\})}$$
- ▶ For binary classification where $y_i \in \{0, 1\}$ and $f_t(x_i) \in \{0, 1\}$
- ▶ This means weights increase for misclassified examples and decrease for correctly classified ones
- ▶ Weights are normalized to form a probability distribution

How do we make predictions with the ensemble?

- ▶ Final prediction uses a weighted majority vote:

$$f_{boost}(x) = \mathbb{1} \left\{ \sum_{t=1}^T \alpha_t f_t(x) \geq \frac{1}{2} \sum_{t=1}^T \alpha_t \right\}$$

- ▶ Each weak learner's vote ($f_t(x) \in \{0, 1\}$) is weighted by its performance (α_t)
- ▶ Models with lower error rates have larger influence on the final prediction
- ▶ The threshold is half the sum of all model weights
- ▶ When the weighted sum favors class 1, we predict 1; otherwise, we predict 0

Algorithm AdaBoost Algorithm

Require: Training data $\{(x_i, y_i)\}_{i=1}^n$, $x \in \mathcal{X}$, $y \in \{0, 1\}$, number of iterations T

Ensure: AdaBoost model

- 1: Initialize weights $w_1(i) = \frac{1}{n}$ for $i = 1 : n$
 - 2: **for** $t = 1$ to T **do**
 - 3: Train classifier f_t on weighted training data with weights w_t
 - 4: Calculate weighted error: $\epsilon_t = \sum_{i=1}^n w_t(i) \mathbb{1}\{y_i \neq f_t(x_i)\}$
 - 5: Calculate model weight: $\alpha_t = \frac{1}{2} \ln \left(\frac{1-\epsilon_t}{\epsilon_t} \right)$
 - 6: Scale weights: $\hat{w}_{t+1}(i) = w_t(i) \cdot e^{\alpha_t(1-2\mathbb{1}\{y_i=f_t(x_i)\})}$
 - 7: Normalize: $w_{t+1}(i) = \frac{\hat{w}_{t+1}(i)}{\sum_j \hat{w}_{t+1}(j)}$
 - 8: **end for**
 - 9: **return** Classification rule: $f_{boost}(x_0) = \mathbb{1}\{\sum_{t=1}^T \alpha_t f_t(x_0) \geq \frac{1}{2} \sum_{t=1}^T \alpha_t\}$
-

Key calculations for Tree 1:

- Initial weights:

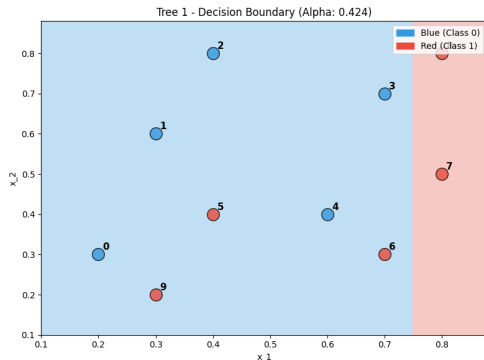
$$w_1(i) = 0.1 \text{ for all samples}$$

- Weighted Error:

$$\epsilon_1 = 0.300$$

- Model Contribution:

$$\begin{aligned}\alpha_1 &= \frac{1}{2} \ln \left(\frac{1 - \epsilon_1}{\epsilon_1} \right) \\ &= \frac{1}{2} \ln \left(\frac{0.7}{0.3} \right) \\ &= 0.424\end{aligned}$$



Weight Update Process:

- For misclassified:

$$w_2(i) \propto w_1(i) \cdot e^{+\alpha_1}$$

$$= 0.1 \cdot e^{0.424}$$

- For correctly classified:

$$w_2(i) \propto w_1(i) \cdot e^{-\alpha_1}$$

$$= 0.1 \cdot e^{-0.424}$$

- Weights are then normalized.

Sample	x_1	x_2	y	w_1	$f_1(x)$	w_2
0	0.2	0.3	0	0.1	0	0.071
1	0.3	0.6	0	0.1	0	0.071
2	0.4	0.8	0	0.1	0	0.071
3	0.7	0.7	0	0.1	0	0.071
4	0.6	0.4	0	0.1	0	0.071
5	0.4	0.4	1	0.1	0	0.167
6	0.7	0.3	1	0.1	0	0.167
7	0.8	0.5	1	0.1	1	0.071
8	0.8	0.8	1	0.1	1	0.071
9	0.3	0.2	1	0.1	0	0.167

Key calculations for Tree 2:

- ▶ Initial weights:

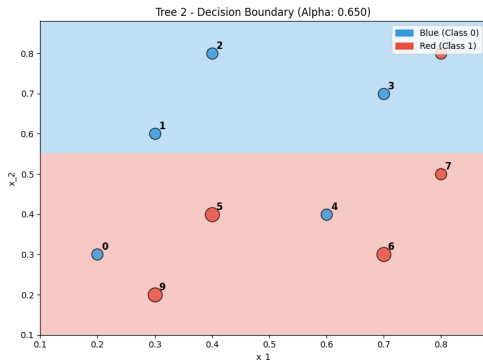
$w_2(i)$ from previous iteration

- ▶ Weighted Error:

$$\epsilon_2 = 0.214$$

- ▶ Model Contribution:

$$\begin{aligned}\alpha_2 &= \frac{1}{2} \ln \left(\frac{1 - \epsilon_2}{\epsilon_2} \right) \\ &= \frac{1}{2} \ln \left(\frac{0.786}{0.214} \right) \\ &= 0.650\end{aligned}$$



Weight Update Process:

- For misclassified:

$$w_3(i) \propto w_2(i) \cdot e^{+\alpha_2}$$

$$= w_2(i) \cdot e^{0.650}$$

- For correctly classified:

$$w_3(i) \propto w_2(i) \cdot e^{-\alpha_2}$$

$$= w_2(i) \cdot e^{-0.650}$$

- Weights are then normalized.

Sample	x_1	x_2	y	w_2	$f_2(x)$	w_3
0	0.2	0.3	0	0.071	1	0.166
1	0.3	0.6	0	0.071	0	0.045
2	0.4	0.8	0	0.071	0	0.045
3	0.7	0.7	0	0.071	0	0.045
4	0.6	0.4	0	0.071	1	0.166
5	0.4	0.4	1	0.166	1	0.106
6	0.7	0.3	1	0.166	1	0.106
7	0.8	0.5	1	0.071	1	0.045
8	0.8	0.8	1	0.071	0	0.166
9	0.3	0.2	1	0.166	1	0.106

Key calculations for Tree 3:

- ▶ Initial weights:

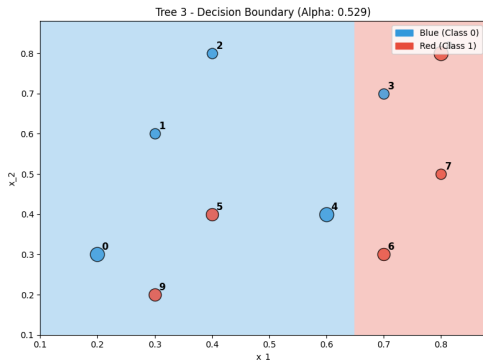
$w_3(i)$ from previous iteration

- ▶ Weighted Error:

$$\epsilon_3 = 0.258$$

- ▶ Model Contribution:

$$\begin{aligned}\alpha_3 &= \frac{1}{2} \ln \left(\frac{1 - \epsilon_3}{\epsilon_3} \right) \\ &= \frac{1}{2} \ln \left(\frac{0.742}{0.258} \right) \\ &= 0.529\end{aligned}$$



Weight Update Process:

- For misclassified:

$$w_4(i) \propto w_3(i) \cdot e^{+\alpha_3}$$

$$= w_3(i) \cdot e^{0.529}$$

- For correctly classified:

$$w_4(i) \propto w_3(i) \cdot e^{-\alpha_3}$$

$$= w_3(i) \cdot e^{-0.529}$$

- Weights are then normalized.

Sample	x_1	x_2	y	w_3	$f_3(x)$	w_4
0	0.2	0.3	0	0.166	0	0.112
1	0.3	0.6	0	0.045	0	0.031
2	0.4	0.8	0	0.045	0	0.031
3	0.7	0.7	0	0.045	1	0.088
4	0.6	0.4	0	0.166	0	0.112
5	0.4	0.4	1	0.106	0	0.206
6	0.7	0.3	1	0.106	1	0.071
7	0.8	0.5	1	0.045	1	0.031
8	0.8	0.8	1	0.166	1	0.112
9	0.3	0.2	1	0.106	0	0.206

Key calculations for Tree 4:

- ▶ Initial weights:

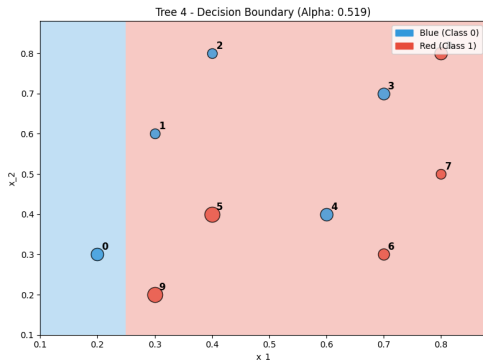
$w_4(i)$ from previous iteration

- ▶ Weighted Error:

$$\epsilon_4 = 0.262$$

- ▶ Model Contribution:

$$\begin{aligned}\alpha_4 &= \frac{1}{2} \ln \left(\frac{1 - \epsilon_4}{\epsilon_4} \right) \\ &= \frac{1}{2} \ln \left(\frac{0.738}{0.262} \right) \\ &= 0.519\end{aligned}$$



Weight Update Process:

- ▶ For misclassified:

$$w_5(i) \propto w_4(i) \cdot e^{+\alpha_4}$$

$$= w_4(i) \cdot e^{0.519}$$

- ▶ For correctly classified:

$$w_5(i) \propto w_4(i) \cdot e^{-\alpha_4}$$

$$= w_4(i) \cdot e^{-0.519}$$

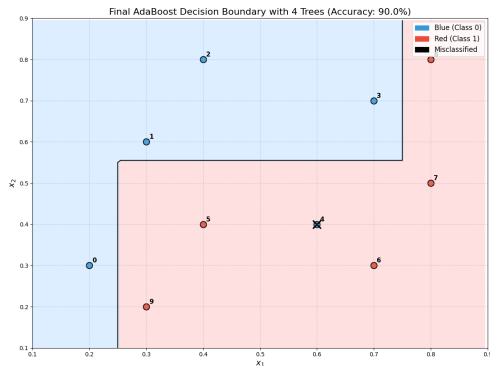
- ▶ Weights are then normalized.

Sample	x_1	x_2	y	w_4	$f_4(x)$	w_5
0	0.2	0.3	0	0.112	0	0.076
1	0.3	0.6	0	0.031	1	0.058
2	0.4	0.8	0	0.031	1	0.058
3	0.7	0.7	0	0.088	1	0.169
4	0.6	0.4	0	0.112	1	0.214
5	0.4	0.4	1	0.206	1	0.139
6	0.7	0.3	1	0.071	1	0.048
7	0.8	0.5	1	0.031	1	0.021
8	0.8	0.8	1	0.112	1	0.076
9	0.3	0.2	1	0.206	1	0.139

Sample	y	f_1	α_1	f_2	α_2	f_3	α_3	f_4	α_4	f_{boost}
0	0	0	0.424	1	0.650	0	0.529	0	0.519	0
1	0	0	0.424	0	0.650	0	0.529	1	0.519	0
2	0	0	0.424	0	0.650	0	0.529	1	0.519	0
3	0	0	0.424	0	0.650	1	0.529	1	0.519	0
4	0	0	0.424	1	0.650	0	0.529	1	0.519	1
5	1	0	0.424	1	0.650	0	0.529	1	0.519	1
6	1	0	0.424	1	0.650	1	0.529	1	0.519	1
7	1	1	0.424	1	0.650	1	0.529	1	0.519	1
8	1	1	0.424	0	0.650	1	0.529	1	0.519	1
9	1	0	0.424	1	0.650	0	0.529	1	0.519	1

The Power of Ensemble Learning

- ▶ The final decision boundary combines all four weak classifiers
- ▶ This demonstrates how AdaBoost transforms simple models into sophisticated classifiers
- ▶ Accuracy improves from 70% (individual trees) to 90% (ensemble)



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Algorithm Gradient Boosting Algorithm

Require: Training data $\{(x_i, y_i)\}_{i=1}^n$, loss function L , number of trees M , learning rate η

Ensure: Gradient Boosted model

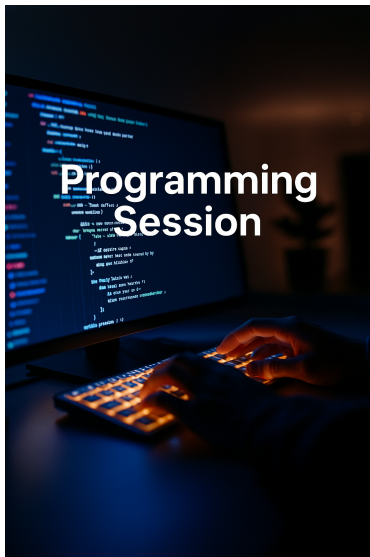
- 1: Initialize model with a constant: $F_0(x) = \arg \min_{\hat{y}} \sum_{i=1}^n L(y_i, \hat{y})$
- 2: **for** $m = 1$ to M **do**
- 3: Compute pseudo-residuals:

$$r_{im} = - \left[\frac{\partial L(y_i, F(x_i))}{\partial F(x_i)} \right]_{F(x)=F_{m-1}(x)} \quad \text{for } i = 1 : n$$
- 4: Fit a new regression tree to the pseudo-residuals r_{im}
- 5: Compute optimal value for each leaf region R_{jm} :

$$\gamma_{jm} = \arg \min_{\gamma} \sum_{x_i \in R_{jm}} L(y_i, F_{m-1}(x_i) + \gamma)$$
- 6: Update equation: $F_m(x) = F_{m-1}(x) + \eta \cdot \gamma_m$
- 7: **end for**
- 8: **return** Final model $F_M(x)$

Feedback Poll

Click here to participate in the poll



Programming Session 4: Introducing Supervised Learning Algorithms for Time Series Forecasting

- ▶ Section 1: Preprocessing the Dataset.
- ▶ Section 2: Tree based Models for Time Series Forecasting.
- ▶ *Click here to access the programming session*

Solution will be posted tonight on the GitHub page.

- ▶ *Click here to access the GitHub Page*

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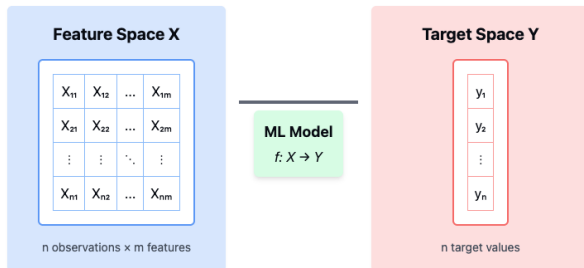
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Supervised Learning:

- ▶ Learn a function $f : X \rightarrow Y$ from labeled data.
- ▶ **Feature space** X : matrix of features, n observations $\times m$ features.
- ▶ **Target space** Y : vector of n labels (or target values).



- ▶ **Feature vector:**

$$\mathbf{x}_i = (x_{i1}, \dots, x_{im}) \in \mathbb{R}^m$$

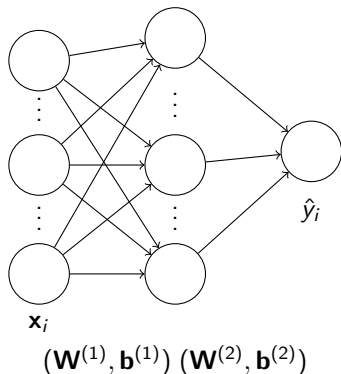
- ▶ **Parameters:**

$$\theta = \{(\mathbf{W}^{(1)}, \mathbf{b}^{(1)}), (\mathbf{W}^{(2)}, \mathbf{b}^{(2)})\}$$

- ▶ $\mathbf{W}^{(1)} \in \mathbb{R}^{m \times L}$: weights connecting input to hidden layer
- ▶ $\mathbf{b}^{(1)} \in \mathbb{R}^L$: biases for hidden layer
- ▶ $\mathbf{W}^{(2)} \in \mathbb{R}^{L \times 1}$: weights connecting hidden to output
- ▶ $\mathbf{b}^{(2)} \in \mathbb{R}$: bias for output layer

- ▶ **Model:** $f_\theta : \mathbb{R}^m \rightarrow \mathbb{R}$ mapping features to predictions

- ▶ **Output:** $\hat{y}_i$ (category for classification or continuous value for regression)



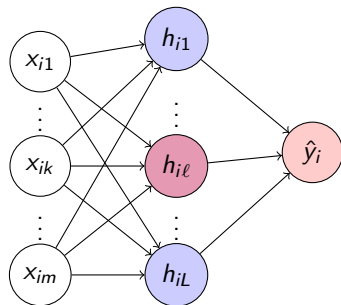
- ▶ For all $\ell \in \{1, 2, \dots, L\}$:

$$h_{i\ell} = \sigma_1 \left(\sum_{k=1}^m W_{k\ell}^{(1)} \cdot x_{ik} + b_{\ell}^{(1)} \right)$$

- ▶ Then the output:

$$\hat{y}_i = \sigma_2 \left(\sum_{\ell=1}^L W_{\ell}^{(2)} \cdot h_{i\ell} + b^{(2)} \right)$$

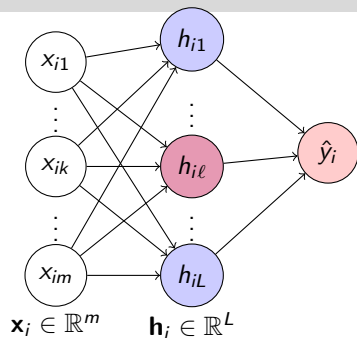
- ▶ σ_1, σ_2 are activation functions.
- ▶ Parameters:
 $\theta = \{(\mathbf{W}^{(1)}, \mathbf{b}^{(1)}), (\mathbf{W}^{(2)}, \mathbf{b}^{(2)})\}$



- ▶ $\mathbf{x}_i \in \mathbb{R}^m$ is the input feature vector
- ▶ $\mathbf{h}_i \in \mathbb{R}^L$ is the hidden layer vector
- ▶ The model output $\hat{y}_i \in \mathbb{R}$ is calculated as follows:

$$\mathbf{h}_i = \sigma_1(\mathbf{W}^{(1)T} \mathbf{x}_i + \mathbf{b}^{(1)})$$

$$\hat{y}_i = \sigma_2((\mathbf{W}^{(2)})^T \mathbf{h}_i + b^{(2)})$$



- ▶ The final output is:

$$\hat{y}_i = f_{\theta}(x_i) = \sigma_2 \left((\mathbf{W}^{(2)})^T \sigma_1 \left(\mathbf{W}^{(1)T} \mathbf{x}_i + \mathbf{b}^{(1)} \right) + b^{(2)} \right)$$

- ▶ **Next step:** Learning the parameters $\theta = \{(\mathbf{W}^{(1)}, \mathbf{b}^{(1)}), (\mathbf{W}^{(2)}, \mathbf{b}^{(2)})\}$ from training data using a loss function

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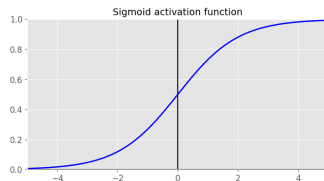
Learning the parameters using Gradient Descent

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Sigmoid:

$$\sigma(z) = \frac{1}{1 + e^{-z}} \in [0, 1]$$

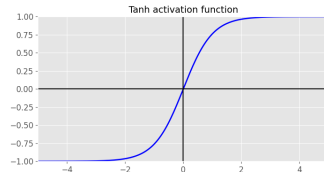
- ▶ Used as final activation for binary classification
- ▶ Output interpreted as probability:
 $\hat{y}_i = P(Y = 1 | \mathbf{x}_i)$



Tanh:

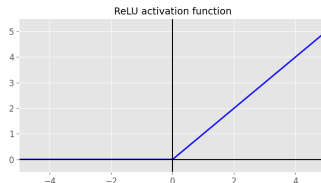
$$\tanh(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}} \in [-1, 1]$$

- ▶ Zero-centered, helps with convergence
- ▶ Often used in hidden layers



ReLU (Rectified Linear Unit):

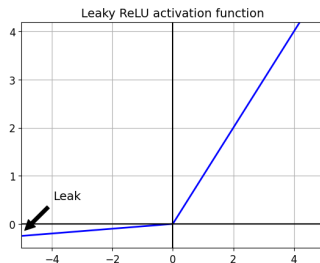
$$\text{ReLU}(z) = \max(0, z)$$



- ▶ Faster to compute, computational efficiency
- ▶ No saturation for positive values, helps with gradient flow
- ▶ Problem: "dying ReLU" (neurons can get stuck at 0)
- ▶ Most widely used in hidden layers
- ▶ Sparse activation: typically 50% of neurons inactive
- ▶ No vanishing gradient for positive inputs

Leaky ReLU:

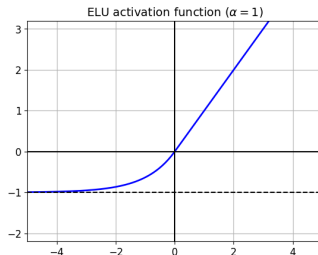
$$\text{LeakyReLU}(z) = \begin{cases} z & \text{if } z > 0 \\ \alpha z & \text{if } z \leq 0 \end{cases}$$



- ▶ Introduced in [4].
- ▶ Prevents "dying ReLU" problem with small slope α
- ▶ Typical values for α range from 0.01 to 0.2
- ▶ Allows small gradient flow for negative inputs
- ▶ Maintains most of the computational efficiency of ReLU
- ▶ Not always superior to ReLU in practice

ELU (Exponential Linear Unit):

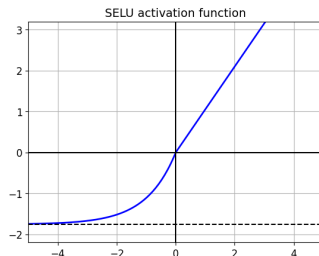
$$\text{ELU}(z) = \begin{cases} z & \text{if } z > 0 \\ \alpha(e^z - 1) & \text{if } z \leq 0 \end{cases}$$



- ▶ Introduced in [1].
- ▶ Smooth curve for negative values, reducing noise
- ▶ Approaches $-\alpha$ as z becomes very negative
- ▶ Self-regularizing: can help with internal covariate shift
- ▶ More computationally expensive than ReLU
- ▶ Often produces faster convergence in training

SELU (Scaled ELU):

$$\text{SELU}(z) = \lambda \begin{cases} z & \text{if } z > 0 \\ \alpha(e^z - 1) & \text{if } z \leq 0 \end{cases}$$



- ▶ Introduced in [3].
- ▶ Fixed parameters: $\alpha \approx 1.67$ and $\lambda \approx 1.05$
- ▶ Scaling factor λ enables self-normalization
- ▶ Automatically preserves mean and variance of inputs
- ▶ Helps training deep networks without batch normalization
- ▶ Requires "SELU initialization" (LeCun Normal)

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► Why do we need loss functions?

- Quantify how well our model f_θ performs on data
- Provide a differentiable objective to optimize
- Guide the learning of parameters θ

► From predictions to learning:

- Forward propagation gives us: $\hat{y}_i = f_\theta(\mathbf{x}_i)$
- Loss function measures: $\mathcal{L}(\hat{y}_i, y_i)$ - the discrepancy between predictions and true values

► Overall objective:

$$\min_{\theta} \frac{1}{n} \sum_{i=1}^n \mathcal{L}(f_\theta(\mathbf{x}_i), y_i)$$

- ▶ **Dataset:** $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^n$ where $y_i \in \{0, 1\}$
- ▶ **Probabilistic view:** Neural network outputs represent probabilities
- ▶ **For binary classification:**
 $\forall i \in \{1, \dots, n\} \quad \hat{y}_i = f_{\theta}(\mathbf{x}_i) = p_{\theta}(Y = 1 | \mathbf{x}_i)$
- ▶ **Log-likelihood for all data:**

$$\begin{aligned}\log \mathcal{L}(\theta) &= \sum_{i=1}^n \log p_{\theta}(Y = y_i | \mathbf{x}_i) \\ &= \sum_{i=1}^n [y_i \log \hat{y}_i + (1 - y_i) \log(1 - \hat{y}_i)]\end{aligned}$$

- ▶ **The loss function is the normalized negative Log-Likelihood:**

$$\min_{\theta} -\frac{1}{n} \log \mathcal{L}(\theta) \iff \min_{\theta} -\frac{1}{n} \sum_{i=1}^n [y_i \log \hat{y}_i + (1 - y_i) \log(1 - \hat{y}_i)]$$

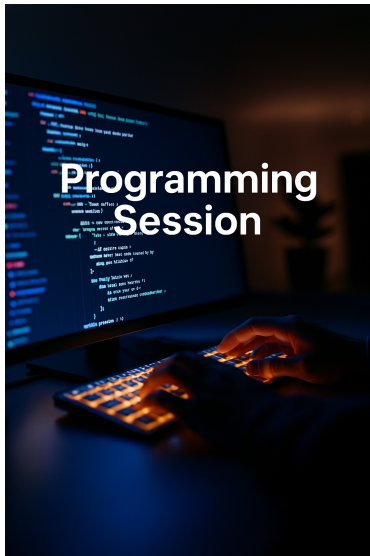
- ▶ **Dataset:** $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^n$ where $y_i \in \{1, 2, \dots, K\}$
- ▶ **Probabilistic view:** Neural network outputs represent probability distribution over K classes
- ▶ **For multi-class classification:**
 $\forall i \in \{1, \dots, n\} \quad \hat{\mathbf{y}}_i = f_{\theta}(\mathbf{x}_i) = (p_{\theta}(Y = 1|\mathbf{x}_i), \dots, p_{\theta}(Y = K|\mathbf{x}_i))$
- ▶ **One-hot encoding of target:** $\mathbf{y}_i = (y_{i1}, y_{i2}, \dots, y_{iK})$ where $y_{ik} = \mathbb{1}_{[y_i=k]}$
- ▶ **Log-likelihood for all data:**

$$\log \mathcal{L}(\theta) = \sum_{i=1}^n \log p_{\theta}(Y = y_i|\mathbf{x}_i) = \sum_{i=1}^n \sum_{k=1}^K y_{ik} \log \hat{y}_{ik}$$

- ▶ **The loss function is the normalized negative Log-Likelihood:**

$$\min_{\theta} -\frac{1}{n} \log \mathcal{L}(\theta) \iff \min_{\theta} -\frac{1}{n} \sum_{i=1}^n \sum_{k=1}^K y_{ik} \log \hat{y}_{ik}$$

Loss	Formula	Key Properties
MSE	$\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$	▶ Differentiable everywhere ▶ Sensitive to outliers
MAE	$\frac{1}{n} \sum_{i=1}^n y_i - \hat{y}_i $	▶ Less sensitive to outliers ▶ Non-differentiable at zero
MAPE	$\frac{1}{n} \sum_{i=1}^n \frac{ y_i - \hat{y}_i }{ y_i }$	▶ Scale-independent



Optional Programming Session: The Custom Huber Loss function

- ▶ *Click here to access the programming session*

Content:

- ▶ Tensors and operations in TensorFlow.
- ▶ Computing Gradients with Autodiff.
- ▶ Custom Loss Function: The Huber Loss.

- ▶ **For classification:**

- ▶ **Binary classification:** Binary Cross-Entropy with sigmoid

$$\mathcal{L}_{BCE} = -\frac{1}{n} \sum_{i=1}^n [y_i \log \hat{y}_i + (1 - y_i) \log(1 - \hat{y}_i)]$$

- ▶ **Multi-class classification:** Categorical Cross-Entropy with softmax

$$\mathcal{L}_{CCE} = -\frac{1}{n} \sum_{i=1}^n \sum_{k=1}^K y_{ik} \log \hat{y}_{ik}$$

- ▶ **Multi-label classification:** Binary Cross-Entropy per label

$$\mathcal{L}_{ML} = -\frac{1}{n} \sum_{i=1}^n \sum_{k=1}^K [y_{ik} \log \hat{y}_{ik} + (1 - y_{ik}) \log(1 - \hat{y}_{ik})]$$

- ▶ **For regression:**

- ▶ **Clean data, normal distribution:** MSE

$$\mathcal{L}_{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

- ▶ **Data with outliers:** MAE

$$\mathcal{L}_{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

- ▶ **Data with outliers (alternative):** Huber Loss

$$\mathcal{L}_{\delta} = \frac{1}{n} \sum_{i=1}^n \begin{cases} \frac{1}{2}(y_i - \hat{y}_i)^2 & \text{if } |y_i - \hat{y}_i| \leq \delta \\ \delta|y_i - \hat{y}_i| - \frac{1}{2}\delta^2 & \text{otherwise} \end{cases}$$

- ▶ **Relative error important:** MAPE

$$\mathcal{L}_{MAPE} = \frac{1}{n} \sum_{i=1}^n \frac{|y_i - \hat{y}_i|}{|y_i|} \times 100\%$$

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Algorithm Gradient Descent Algorithm

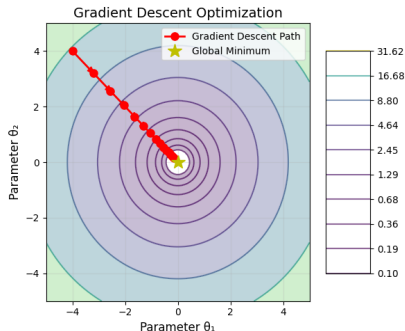
Require: Training data $\{(\mathbf{x}_i, y_i)\}_{i=1}^n$, loss function $\mathcal{L}$, learning rate α , iterations T

Ensure: Optimized parameters θ

- 1: Initialize parameters $\theta^{(0)}$ randomly
 - 2: **for** $t = 1$ to T **do**
 $\theta^{(t)} = \theta^{(t-1)} - \alpha \cdot \nabla_{\theta} \mathcal{L}(\theta^{(t-1)})$
 - 3: **if** Convergence criteria met **then**
 - 4: **break**
 - 5: **end if**
 - 6: **end for**
 - 7: **return** Final parameters $\theta^{(T)}$
-

Key aspects of Gradient Descent:

- ▶ **Intuition:** Move downhill in the direction of steepest descent
- ▶ **Learning rate** α controls step size:
 - ▶ Too small: slow convergence
 - ▶ Too large: overshooting or divergence
- ▶ For neural networks, loss landscapes are typically **non-convex**, so we often converge to a **local minimum**



- ▶ In this figure, we reach the **global minimum** since the function is **convex**

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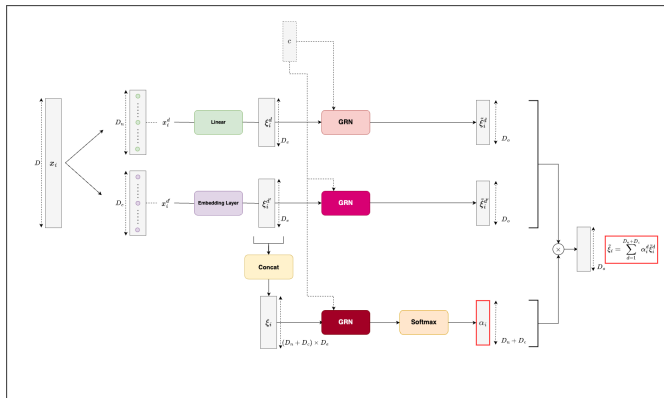
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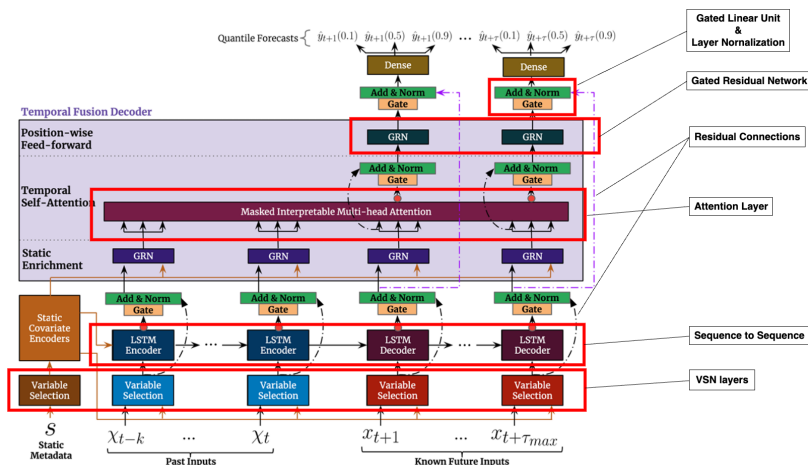
Learning the parameters using Gradient Descent

Introducing the Variable Selection Network

- We will explore the Variable Selection Network (VSN) in Lecture 7 as part of the Temporal Fusion Transformer Architecture.

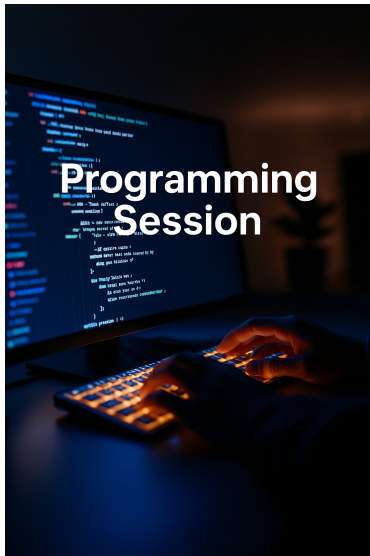


TFT Architecture: High-Level View



Feedback Poll

Click here to participate in the poll



Programming Session 4: Introducing Supervised Learning Algorithms for Time Series Forecasting

- ▶ Section 3: Neural Networks.
- ▶ Section 4: Performance Analysis.
- ▶ *Click here to access the programming session*

Solution will be posted tonight on the GitHub page.

- ▶ *Click here to access the GitHub Page*

Quiz Time!

Click here to take the quiz

Thank you for your attention

- [1] Djork-Arné Clevert, Thomas Unterthiner, and Sepp Hochreiter. “Fast and accurate deep network learning by exponential linear units (elus)”. In: *arXiv preprint arXiv:1511.07289* (2015).
- [2] Yoav Freund, Robert E Schapire, et al. “Experiments with a new boosting algorithm”. In: *icml*. Vol. 96. Citeseer. 1996, pp. 148–156.
- [3] Günter Klambauer et al. “Self-normalizing neural networks”. In: *Advances in neural information processing systems* 30 (2017).
- [4] Bing Xu et al. “Empirical evaluation of rectified activations in convolutional network”. In: *arXiv preprint arXiv:1505.00853* (2015).